

Volume 5: Tournament Platform & Influencer Group Battles

Technical specification for solo knockout tournaments, influencer team battles, cryptographic gate code generation, GMT start-time voting tallies, automated match scheduling, and lobby experiences.

Target Stack: NestJS 11 · Prisma 6 · BullMQ · Redis · Socket.IO · Lead: Muhammad Sameer Ali

VOLUME OBJECTIVE:
Specifies the complete tournament lifecycle state machine. Covers regular solo brackets, influencer-led team battles (2–6 teams), gate codes (sponsorCode, joinCode, captainCode, memberCode), GMT start-time voting, and live arena entry gates.

1. Tournament Types & Influencer Team Rules

- **REGULAR Solo Knockout:** Standard multi-round tournament. Players compete 1v1 until a single champion remains.
- **INFLUENCER Group Battle:** 2 to 6 named teams (groupCount). Each team is led by a named influencer-captain. Influencers invite followers via memberCode. The last surviving player in the tournament wins the title and prize pool for their entire team.

2. Cryptographic Gate Code Engine

Code Field	Target Audience & Usage	Security & Validation Rules
sponsorCode	Issued to sponsor entity to claim tournament ownership.	Unique 8-char string. Unlocks sponsor studio controls.
joinCode	Distributed to players for private tournaments.	Required payload in POST /promo-tournaments/:id/join.
captainCode	Issued to influencer to join as Team Captain.	Grants special Captain card styling and team ownership in roster.
memberCode	Shared by captain with followers to join their team.	Assigns player to corresponding TournamentTeam.

3. GMT Start-Time Voting & Automated Scheduling

- **Admin Setup:** Admin picks startDate (GMT date) and candidate slots timeOptions (e.g. `[14:00, 18:00, 21:00]`).
- **Player Voting:** Registered players cast votes via POST /promo-tournaments/:id/vote-time.
- **Auto-Scheduler Cron:** 2 hours before target date, BullMQ job locks votes, selects winning slot, sets `startAt`, and sends WebSocket alerts.

4. Tournament Lobby Experience (`/events/[id]`)

The lobby card displays live countdown to `startAt`, sponsor banner, charity cause goal, registered players list, and active team rosters. When match starts, transitions directly into `CountDown31.tsx`.